

Ex1b: HMM Based Predictive Text System

PROGRAM:

```
import nltk
from nltk.corpus import brown
from nltk.tag import hmm
from collections import Counter, defaultdict

nltk.download('brown')
nltk.download('universal_tagset')

sentences = brown.sents()

tokens = [word.lower() for sent in sentences for word in sent]
print("Total number of tokens:", len(tokens))

tagged_sentences = brown.tagged_sents(tagset='universal')
tagged_sentences = [
    [(word.lower(), tag) for (word, tag) in sent]
    for sent in tagged_sentences
]

trainer = hmm.HiddenMarkovModelTrainer()
hmm_model = trainer.train_supervised(tagged_sentences)

transition_probs = defaultdict(Counter)
for sent in tagged_sentences:
    tags = [tag for (_, tag) in sent]
    for i in range(len(tags) - 1):
        transition_probs[tags[i]][tags[i + 1]] += 1

for tag in transition_probs:
    total = sum(transition_probs[tag].values())
    for next_tag in transition_probs[tag]:
        transition_probs[tag][next_tag] /= total

emission_probs = defaultdict(Counter)
for sent in tagged_sentences:
    for word, tag in sent:
        emission_probs[tag][word] += 1

for tag in emission_probs:
    total = sum(emission_probs[tag].values())
    for word in emission_probs[tag]:
        emission_probs[tag][word] /= total

def predict_next_word(input_words):
    input_words = [w.lower() for w in input_words]
    predicted_tags = hmm_model.tag(input_words)
    last_tag = predicted_tags[-1][1]
    next_pos = max(transition_probs[last_tag], key=transition_probs[last_tag].get)
    next_word = max(emission_probs[next_pos], key=emission_probs[next_pos].get)
    return next_word, next_pos
```

```
test_sentence = ["the", "government"]
print("Input Sentence:", test_sentence)
print("Predicted Next Word and POS:", predict_next_word(test_sentence))
print("Predicted POS Tags:", hmm_model.tag(test_sentence))
```

OUTPUT:

```
Total number of tokens: 1161192
Input Sentence: ['the', 'government']
Predicted Next Word and POS: ('', '.')
```

Word	POS Tag
the	DET
government	NOUN

```
Predicted POS Tags: [('the', 'DET'), ('government', 'NOUN')]
```